

Homework 2

You're only allowed to use the standard built-in Python 3 libraries with the exception of Numpy and Matplotlib.

Question 1. Plot the following trigonometric functions for $-5\pi \leq x \leq 5\pi$.

$$f(x) = x \sin^2(x), \quad g(x) = -x \sin^2(x)$$

- The title of the plot should read "Trigonometric Functions."
- The x-axis label of the plot should read "x-axis."
- The y-axis label of the plot should read "y-axis."
- The legends should appropriately denote the different functions by their definition.
- The graphs should have smooth continuous curves.

Question 2. Download and save the novel "**The Cosmic Computer**" by Piper. Read the saved text file into Python and plot a bar graph of the character frequency (case insensitive).

- The title of the plot should read 'Character Frequency in "The Cosmic Computer" by Piper.'
- The x-axis label of the plot should read "Alphabet."
- The y-axis label of the plot should read "Frequency."
- The bars should be sorted in ascending order.

Question 3. Using `matplotlib.pyplot.imshow` and an eight by eight binary Numpy array, plot the chess board pattern.

- The x-ticks should be letters from A to H, A being at the left.
- The y-ticks should be numbers from 1 to 8, one being at the bottom.
- The title should be "Chess Board Pattern."

Question 4. Following are the two matrices from the last homework,

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 4 & 2 & 2 \end{bmatrix}$$

Use Numpy to multiply them and print the product $\mathbf{AB}$. Does it match with the one you carried out by hand?

Question 5. The Gregory series is given as,

$$f(n) = \sum_{i=1}^{i=n} \frac{4(-1)^{i+1}}{2i-1} = \frac{4}{1} - \frac{4}{3} + \frac{4}{5} - \dots + \frac{4(-1)^{n+1}}{2n-1}$$

1) Without using any loops, i. e., using vectorised operations over Numpy arrays, plot the following values,

$$\{(n, f(n)) : 1 \leq n \leq 100\}$$

2) Plot the following error curve,

$$\{(n, (\pi - f(n))^2) : 1 \leq n \leq 100\}$$

Label the axis and title each plot appropriately. What is $\lim_{n \rightarrow \infty} f(n)$?

Question 6. Consider the following study results in table 1,

State all the points that are,

- 1) True Negatives
- 2) True Positives
- 3) False Positives
- 4) False Negatives

Point	Real	Predicted
1	−	+
2	−	−
3	−	−
4	−	+
5	+	−
6	+	+
7	+	+
8	+	+

TABLE 1—Real and Predicted Labels

Question 7. Give the confusion matrix for table 1.

	Predicted −	Predicted +
Real −	TN	FP
Real +	FN	TP

TABLE 2—Binary Confusion Matrix

Question 8. Again, for table 1, give the following in exact fractions.

- 1) Baseline Accuracy
- 2) Accuracy

$$\frac{\text{Total Correctly Classified}}{\text{Total Examples}}$$

- 3) Precision

$$\frac{\text{True Positives}}{\text{Total Examples Classified as Positive}}$$

- 4) Recall

$$\frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

- 5) F1 Score

$$\frac{2TP}{2TP + FP + FN}$$

Question 9. Often (but not always) how are precision and recall related?

Question 10. Between precision and recall, which is a better metric for,

- 1) Predicting good restaurants.
- 2) Predicting cancer.

Question 11. Between precision and recall, which metric is used in the Receiver Operator's Curve (ROC). What does the area under this curve tell us about a prospective classifier?

SUBMISSION INSTRUCTIONS

- 1) Submit a PDF that answers all the questions and includes any plots that the assignment asks for. Circle and/emphasize your final answer wherever possible.
- 2) Submit a Python file, e. g., plot.py that generates all the plots one by one.